

WEEK 4 – ASSIGNMENT

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Dit University

Title:

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

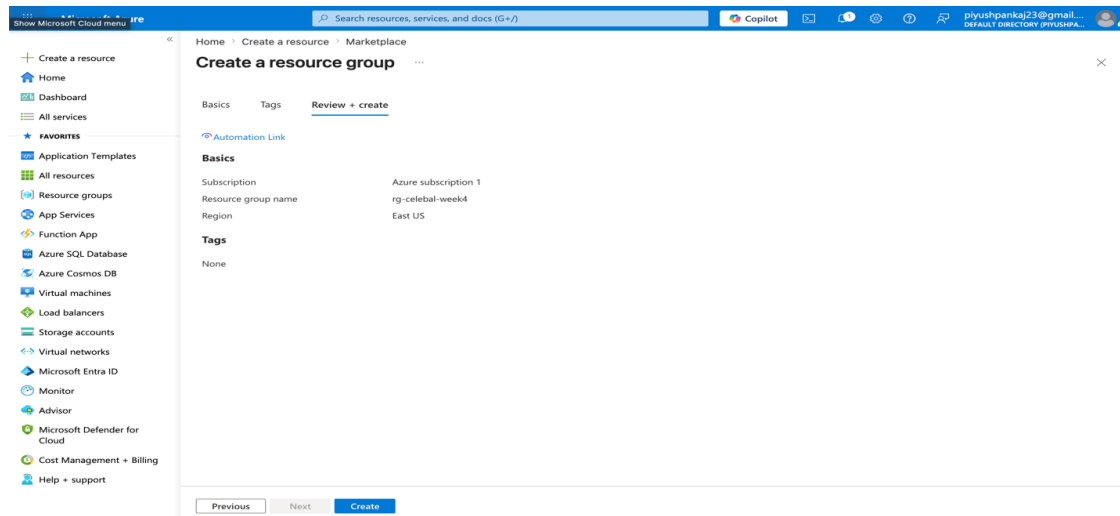
Assignment Tasks

Task 1:

- Explore Azure Portal
- Create a Resource Group

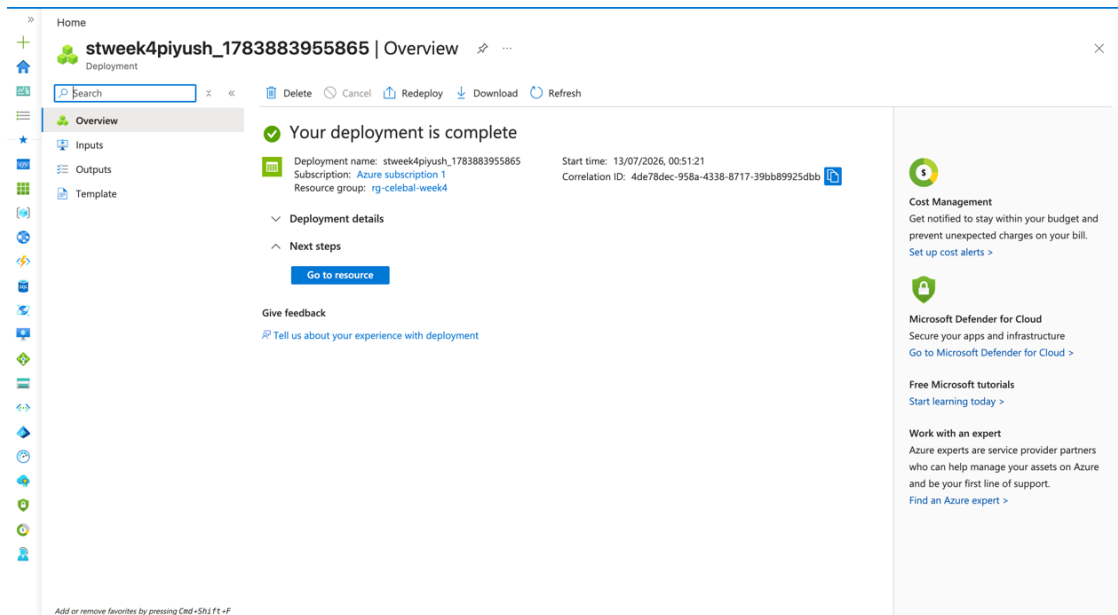
Deliverable:

- Screenshot of Resource Group



Resource group 'rg-celebal-week4' created (Review + Create)

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Resource group overview — showing Data Factory and Storage Account provisioned inside it

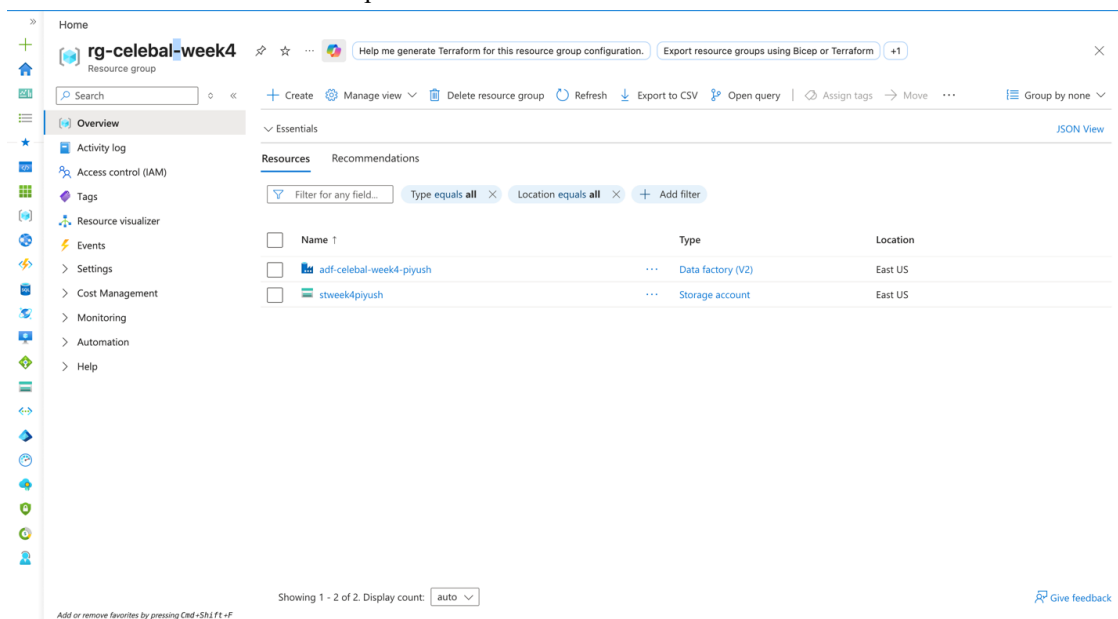
Task 2: Storage Setup

Do:

- Create a Storage Account
- Create a Blob Container
- Upload a CSV file

Deliverable:

- Screenshot of container with uploaded file



Storage account deployment complete

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The screenshot shows the Azure portal interface for a storage account named 'stweek4piyush'. The left sidebar contains navigation links for Overview, Activity log, Tags, Diagnose and solve problems, Access Control (IAM), Data migration, Events, Storage browser, Storage Mover, Partner solutions, Resource visualizer, Data storage, Containers, Classic file shares, Queues, Tables, Security + networking, Data management, Settings, and Monitoring. The main content area is divided into several sections: Essentials (Resource group, Location, Subscription, Subscription ID, Disk state, Tags), Performance (Standard, Locally redundant storage (LRS), StorageV2 (general purpose v2), Succeeded, Created), Properties (Blob service, File service), Monitoring, Capabilities (7), Recommendations (0), Tutorials, and Tools + SDKs. The Blob service section shows various settings like Hierarchical namespace, Default access tier, Blob anonymous access, Blob soft delete, Container soft delete, Versioning, Change feed, NFS v3, Allow cross-tenant replication, and Storage tasks assignments. The File service section shows Identity-based access. The Security section shows Require secure transfer for REST API operations, Storage account key access, Minimum TLS version, and Infrastructure encryption. The Networking section shows Public network access, Public network access scope, Private endpoint connections, Network routing, and Endpoint type.

Storage account overview / properties

The screenshot shows the Azure portal interface for a storage account named 'raw-data'. The left sidebar contains navigation links for Overview, Diagnose and solve problems, Access Control (IAM), and Settings. The main content area is divided into several sections: Overview (Authentication method, Add filter, Search blobs by prefix, Only show active blobs), Showing all 1 items, and a table of blobs. The table has columns for Name, Last modified, Access tier, Blob type, Size, and Lease state. The only blob listed is 'Sample - Superstore.csv' with a size of 2.18 MiB and a lease state of Available.

'raw-data' container with 'Sample - Superstore.csv' uploaded

Task 3: ADF Basics

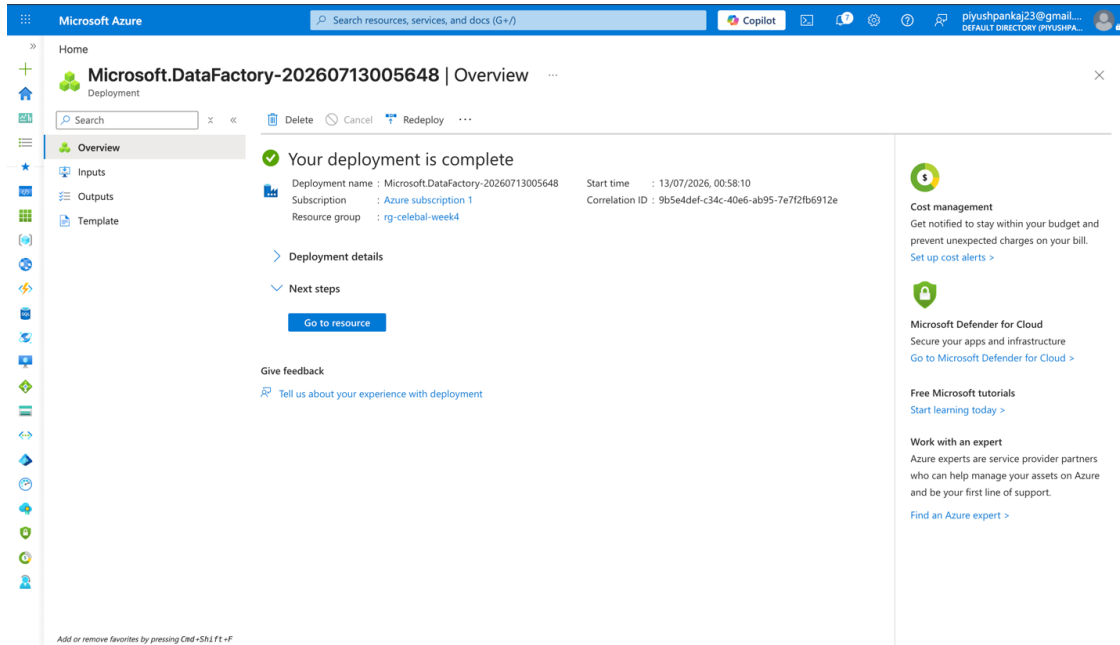
Do:

- Create Azure Data Factory
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)
- Create datasets (source + destination)
- Use Get Metadata activity

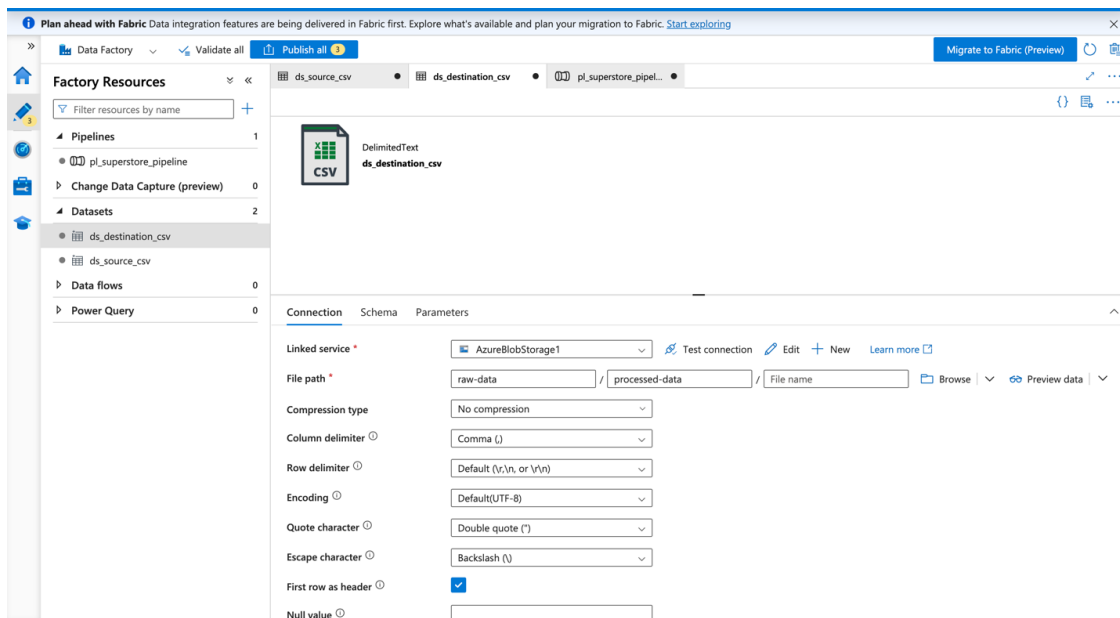
Deliverable:

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- Screenshot of Linked Service
- Screenshot of dataset
- Screenshot of Get Metadata activity

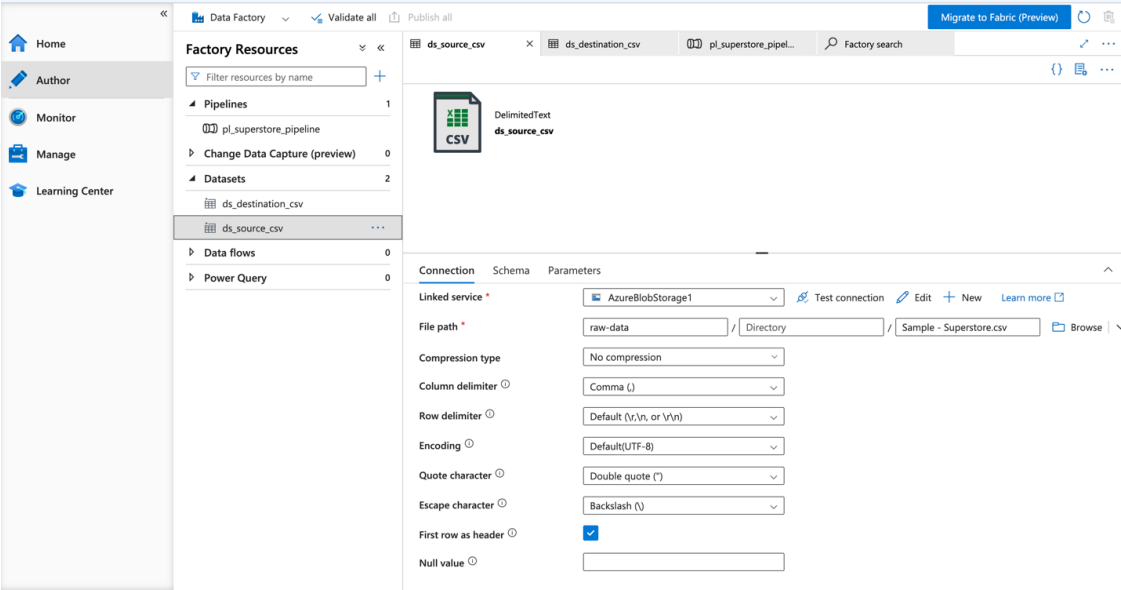


Azure Data Factory deployment complete

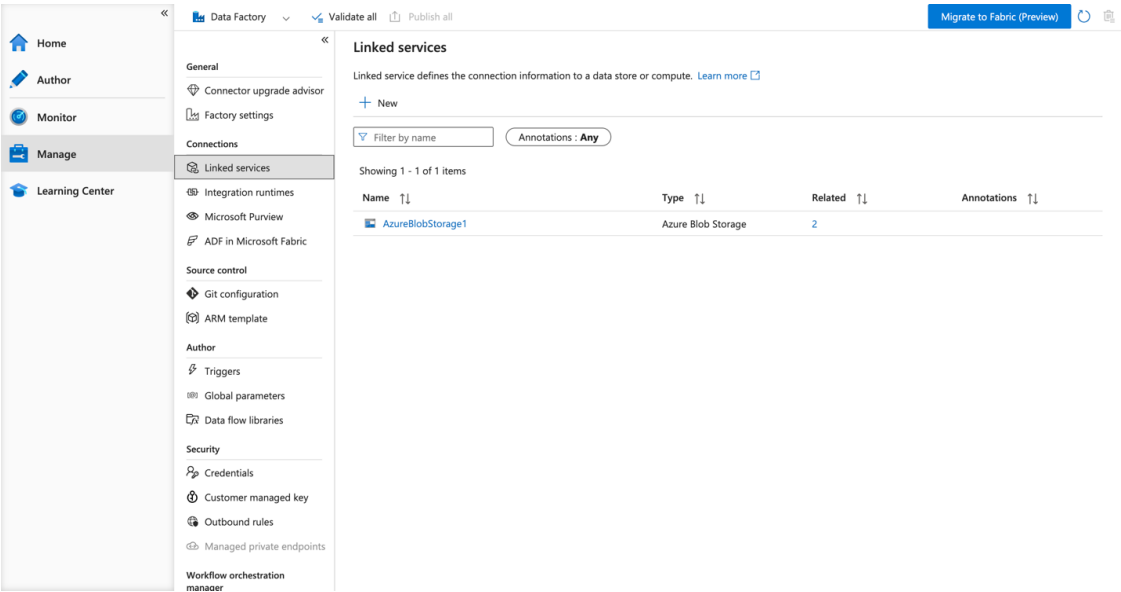


Linked Service 'AzureBlobStorage1' created

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Source dataset 'ds_source_csv'



Destination dataset 'ds_destination_csv'

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Input

 Copy to clipboard

```
{
  "dataset": {
    "referenceName": "ds_source_csv",
    "type": "DatasetReference",
    "parameters": {}
  },
  "fieldList": [
    "itemName",
    "size",
    "lastModified"
  ],
  "storeSettings": {
    "type": "AzureBlobStorageReadSettings",
    "enablePartitionDiscovery": false
  },
  "formatSettings": {
    "type": "DelimitedTextReadSettings"
  }
}
```

Get Metadata activity — field list input (itemName, size, lastModified)

Task 4: Pipeline Development

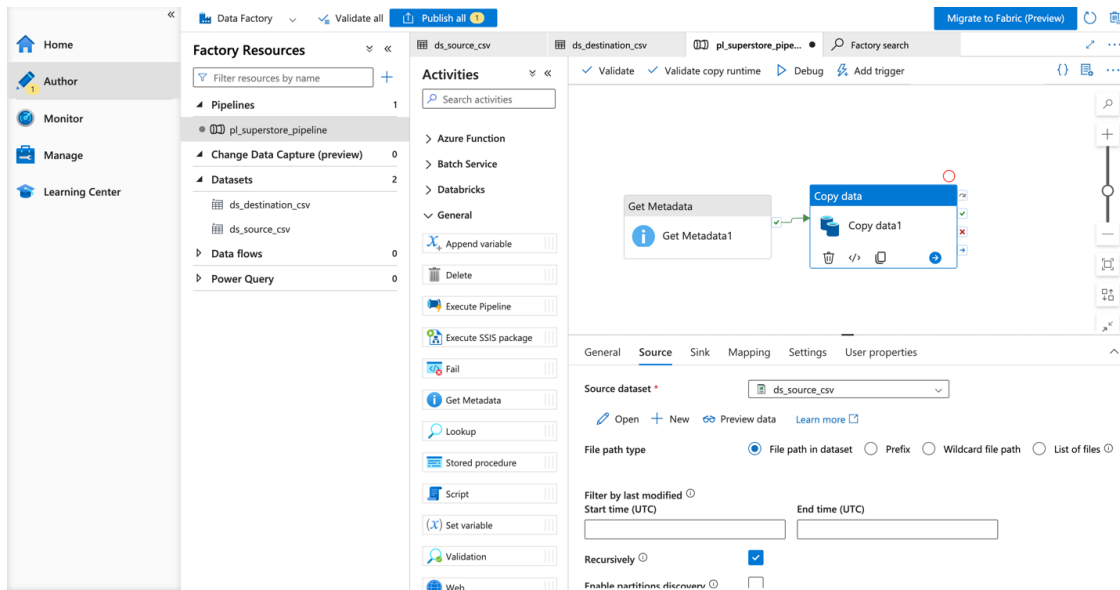
Do:

- Create pipeline using Copy Data activity
- Configure source and destination
- Add ForEach activity (optional if multiple files)

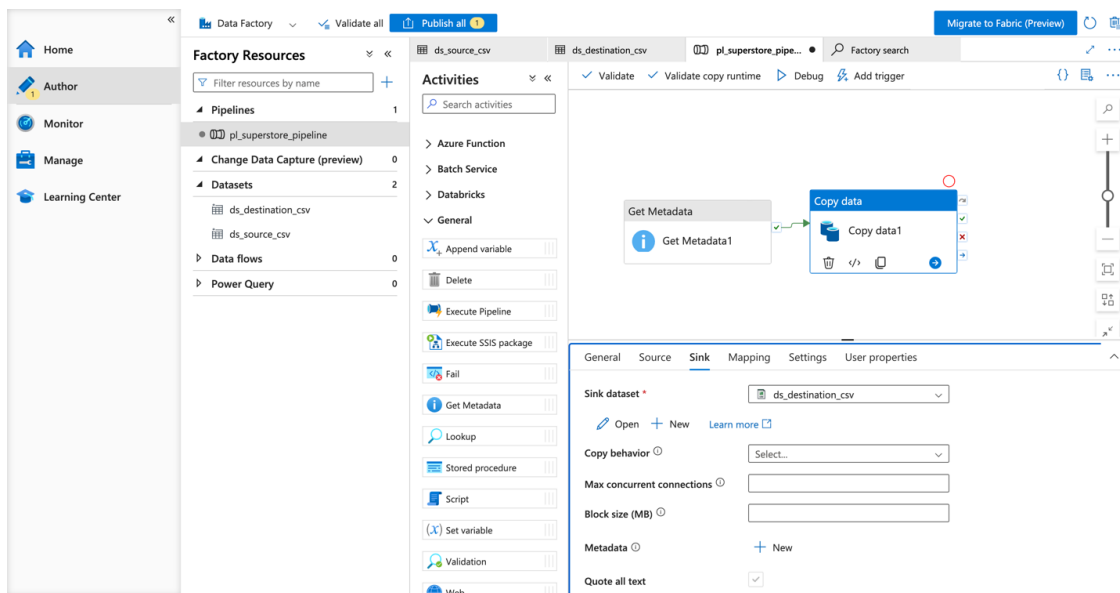
Deliverable:

- Screenshot of pipeline design

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Pipeline design — Get Metadata connected to Copy Data (Source configuration)



Pipeline design — Copy Data Sink configuration

Task 5: Pipeline Execution

Do:

- Run pipeline using Debug/Trigger

Deliverable:

- Screenshot showing pipeline execution (Succeeded)

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Microsoft Azure | Data Factory | adf-celebal-week4-piyush

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Home | Author | Monitor | Manage | Learning Center

Factory Resources

- Pipelines: 1
 - pl_superstore_pipeline
- Datasets: 2
 - ds_destination_csv
 - ds_source_csv
- Data flows: 0
- Power Query: 0

Activities

- Move and transform
- Synapse
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

Get Metadata1 → Copy data1

Pipeline: c410a240-fb17-4c8f-a7ce-ab42349a7dd0
run ID: ce-ab42349a7dd0
Pipeline status: Succeeded
View debug run consumption

All status: Monitor in Azure Metrics | Export to CSV

Showing 1 - 2 of 2 items

Activity name	Activity status	Activity name	Run start	Duration
Copy data1	Succeeded	Copy data	7/13/2026, 1:32:27 AM	24s
Get Metadata1	Succeeded	Get Metadata	7/13/2026, 1:32:09 AM	16s

Pipeline run — both activities Succeeded

Task 6: IAM Roles

Do:

- Assign roles: Reader, Contributor
- Provide access to ADF for Storage

Deliverable:

- Screenshot of role assignment

Home | rg-celebal-week4 | stweek4piyush

stweek4piyush | Access Control (IAM)

Storage account

Search | Add | Download role assignments | Edit columns | Refresh | Delete | Feedback

Check access: Role assignments | Roles | Deny assignments

Number of role assignments for this subscription: 4 / 4000

Search by name or email

Type: All | Role: All | Scope: All scopes | Group by: Role

All (4) | Job function roles (2) | Privileged administrator roles (2)

Name	Type	Role	Scope	Condition
Owner (2)				
Piyush Pankaj	User	Owner	Subscription (Inherited)	None
Piyush Pankaj	User	Owner	Subscription (Inherited)	None
Reader (1)				
adf-celebal-week4-piyush	Managed identity	Reader	This resource	None
Storage Blob Data Contributor (1)				
adf-celebal-week4-piyush	Managed identity	Storage Blob Data Contributor	This resource	Add

Showing 1 - 4 of 4 results.

Role assignments — ADF managed identity granted 'Reader' and 'Storage Blob Data Contributor' on the storage account

Mini Project (Final Task)

Problem Statement:

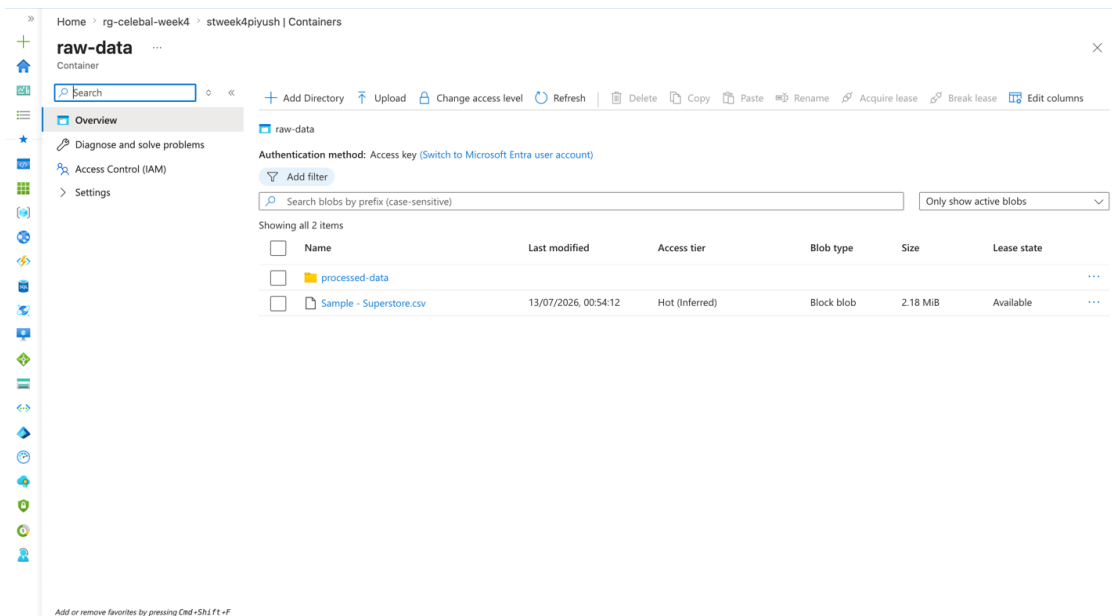
Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

Requirements:

- Source: CSV file in Blob
- Use: Linked Service + Dataset + Pipeline
- Process: Copy Data + Metadata check
- Destination: New file/location

Expected Output:

- Pipeline executed successfully
- Data copied to destination
- Metadata validated



'raw-data' container after pipeline run — 'processed-data' destination folder created alongside the source file

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The screenshot displays the Microsoft Azure Data Factory console. The left sidebar shows the navigation menu with options like Home, Author, Monitor, Manage, and Learning Center. The main area is divided into several sections: Factory Resources, Activities, and a detailed view of the pipeline run.

Factory Resources:

- Pipelines: 1 (pl_superstore_pipeline)
- Change Data Capture (preview): 0
- Datasets: 2 (ds_destination_csv, ds_source_csv)
- Data flows: 0
- Power Query: 0

Activities:

- Move and transform
- Synapse
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

Pipeline Run Details:

- Pipeline: c410a240-fb17-4c8f-a7ce-ab42349a7dd0
- Run ID: ce-ab42349a7dd0
- Pipeline status: Succeeded
- View debug run consumption

Activity Run Details:

Activity name	Activity status	Activity name	Run start	Duration
Copy data1	Succeeded	Copy data	7/13/2026, 1:32:27 AM	24s
Get Metadata1	Succeeded	Get Metadata	7/13/2026, 1:32:09 AM	16s

End-to-end pipeline (Get Metadata → Copy Data) executed successfully — metadata validated and data copied to destination

This mini project reuses the same end-to-end pipeline (pl_superstore_pipeline) built in Tasks 3–5. It reads 'Sample - Superstore.csv' from the 'raw-data' Blob container, validates its metadata (item name, size, last modified) via Get Metadata, and copies it to 'raw-data/processed-data/' via Copy Data. Both activities executed successfully, confirming the pipeline meets all stated requirements and expected outputs.